

Last time

$f: D \rightarrow \mathbb{R}$, $x \in D$, x is a max of f .

$\Rightarrow$ If f differentiable at x : $f'(x) = 0$

- f is not differentiable

- x is an endpoint of D (if D is an interval)

Def $f: D \rightarrow \mathbb{R}$, $x \in D$

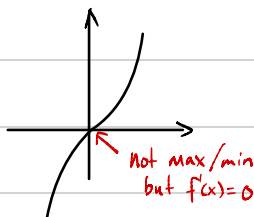
x is a local max if $\exists U \subseteq D$ open containing x s.t. $f|_U: U \rightarrow \mathbb{R}$

x is a maximum for $f|_U$.

Cor If $f: D \rightarrow \mathbb{R}$ & has a local max at x & differentiable, then $f'(x) = 0$.

Rmk The converse is false

EX. $f(x) = x^3$



Def A critical pt. of a function f is a pt. x in the domain of f s.t. f differentiable at x , $f'(x) = 0$

In addition, any pt. in the domain f is not differentiable & any pt. on the boundary of domain are also critical pts.

$\hookrightarrow$ pt $y \in X$ is on the boundary of X if $\forall y \in U$ open neighborhood U contains at least 1 pt. that is not in X . * Topology

Ex ① Find max/min of $f(x) = x^3 - x^2 - 8x + 1$ on $[-2, 2]$

Solu The critical pts. of $f(x)$ are x s.t. $f'(x) = 3x^2 - 2x - 8 = 0$

$$3x^2 - 2x - 8 = (3x + 4)(x - 2)$$

$$x = 2, x = -4/3$$

• f is diff. on $(-2, 2)$

so no critical pts coming from f not being diff.

• pts. on the boundary of $[-2, 2] \Rightarrow x = -2, 2$

• Critical pts. are $-2, -4/3, 2$.

$$f(-4/3) \approx 7.518 \quad \text{global max.}$$

$$f(2) = -11 \quad \text{global min.}$$

$$f(-2) = 5$$

② $f(x) = |x|$ on $[-1, 1]$

Diff. at $(-1, 1) \setminus \{0\}$

No critical pts. s.t. $f'(x) = 0$

$x = 0$ not differentiable $\leftarrow$ min

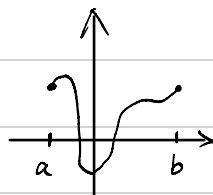
end pts. $-1, 1 \leftarrow$ max

Thm Rolle's Thm: If f cts. on $[a, b]$, diff on (a, b) . & $f(a) = f(b)$

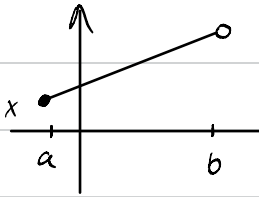
then $x \in (a, b)$ s.t. $f'(x) = 0$

• If f is not diff. the thm doesn't work

Ex $f(x) = |x|$



- If f is not cts. thm doesn't work



Pf. By EVT, f has an max/min on $[a, b]$, say at x

If $x \in (a, b)$. then since x max/min

$$f'(x) = 0, \text{ we're done}$$

If $x \notin (a, b)$, this means that max/min of f happen in the endpts.

By EVT, these definitely exist (at a or b only)

- If $\max = \min = a$ or b

$$\Rightarrow f(x) \text{ constant} \Rightarrow f'(x) = 0$$

WOLG $\max = a, \min = b$

Since we're given $f(a) = f(b)$

$$\text{Still } \max = \min \Rightarrow f \text{ constant} \Rightarrow f'(x) = 0$$

Mean Value Theorem

If f cts. on $[a, b]$, f diff on (a, b) , then $\exists x \in (a, b)$ s.t. $f'(x) = \frac{f(b) - f(a)}{b - a}$.

Pf. Strategy Modify f s.t. it satisfies the conditions of Rolle's Theorem

$$h(x) = f(x) - \left[\frac{f(b) - f(a)}{b - a} \right] (x - a)$$

$$\Rightarrow h(a) = f(a) - \left[\frac{f(b) - f(a)}{b - a} \right] (a - a) = f(a)$$

$$h(b) = f(b) - \left[\frac{f(b) - f(a)}{b - a} \right] (b - a) = f(b) - (f(b) - f(a)) = f(a)$$

$$\Rightarrow h(a) = h(b)$$

h is diff. on (a, b) : Since f is diff. on (a, b) & $\left(\frac{f(b) - f(a)}{b - a} \right) (x - a)$ diff.
So their sum is diff.

By Rolle's Thm, $\exists x \in (a, b)$ s.t. $h'(x) = 0$

$$\text{In addition, } h'(x) = f'(x) - \frac{f(b) - f(a)}{b - a} = 0$$

$$\Rightarrow f'(x) = \frac{f(b) - f(a)}{b - a}$$



Cor If f is defined on $[a, b]$ & $f'(x) = 0 \ \forall x \in (a, b)$. then f is constant.

Pf. Let $\alpha, \beta \in (a, b)$, WTS $f(\alpha) = f(\beta) \ \forall \alpha, \beta$

(then f is constant)

Since f diff. on (α, β) + cts. on $[\alpha, \beta]$

$$\exists x \in (\alpha, \beta), f'(x) = \frac{f(\beta) - f(\alpha)}{\beta - \alpha}$$

$$\text{But } f'(x) = 0 \ \forall x \in (\alpha, \beta) \text{ so } \frac{f(\beta) - f(\alpha)}{\beta - \alpha} = 0$$

$$\Rightarrow f(\beta) - f(\alpha) = 0$$

So f is constant.



Cor If f & g are defined + diff on I , $f'(x) = g'(x)$

$$\Rightarrow f(x) = g(x) + C$$